Scale-rich network metamaterials grow by local rules on a recursive cell division

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(Dated: November 12, 2026)

Scale-rich (SR) metamaterials are planar ligament networks grown by nucleating a segment at a random point and angle and extending it until it meets the existing structure, with a thickness that decays with the time of birth. In the dual graph, where ligaments are nodes and junctions are links, the degree distribution has a k^{-3} tail that was attributed to a spatial preferential attachment. Here we show that, at zero thickness, the SR construction is Cowan's iterated division of cells with area-weighted selection, and that its dual graph grows by a face-local rule: a new node joins the two nodes that carry the sides of one face. Three exact statements follow. The area fractions of the two daughter cells are uniform, so the cell hierarchy is a random binary search tree and the cell areas are uniform spacings. A new ligament ends on a ligament born at $h = tU^2$, with U uniform, so the thickness ratio between a ligament and its host does not depend on time. The degree of an interior ligament equals its number of face incidences, which yields linear preferential attachment and the Barabási–Albert distribution $p_k = 12/[k(k+1)(k+2)]$. Using integral geometry we compute the probability that a new ligament cuts off a corner of its cell ($2\ln 2/\pi$ for a square, 0.448 averaged over cells), which sets the clustering spectrum $C(k) \simeq 2[p_0 + p(k-2)]/[k(k-1)]$. A purely topological face-splitting rule reproduces $P(k)$, $C(k)$ and $k_{nn}(k)$ but not distances; the half of the splits that close squares boosts the shortest-path multiplicity above the configuration-model value. The exponent $\gamma = 3$ is pinned by the ratio of links received to face incidences created; thickness, jamming, angle sets, alignment and density gradients do not move it, whereas coherent orientation fields ($\gamma \simeq 2.4$–2.5 up to $T = 10^6$) and three-dimensional plates ($\gamma \simeq 2.5$) do. At fixed decay exponent, the reported per-ligament energy law cannot distinguish degree from length or age.

I. INTRODUCTION

Both *et al.* [1] introduced scale-rich (SR) mechanical metamaterials, two-dimensional ligament networks in which length, thickness and connectivity span orders of magnitude. The construction has two parameters. At step $t = 1, 2, \ldots, T$ a point is drawn uniformly in the unit square together with a uniform angle, and a ligament of thickness $\lambda_t = \lambda_0 t^{-\alpha}$ is grown both ways from it until each end meets an existing ligament or the boundary. Every junction is a T. Following the road-network convention [2, 3], the authors represent each ligament as a node and each junction as a link. They report power-law distributions of thickness, length and degree, with $P(k) \sim k^{-3}$ attributed to a spatial preferential attachment, because longer ligaments offer more opportunities to be hit. They also report a per-ligament elastic energy law $U_i \sim k_i^{3/2}\lambda_i^{1/2} f(\theta_i)$ and a density scaling of the total energy controlled by α [1].

Two questions are natural from the point of view of network growth. Which growth class does the SR dual graph belong to, and which of its properties are fixed by that class rather than by the design parameters (α, λ_0)? Local growth rules, in which a new node attaches to an existing node and to some of its neighbours, generate preferential attachment without global information, together with a hierarchical clustering spectrum $C(k) \sim 1/k$ and degree correlations [4–7]. Here we show that the SR construction is of this type. The geometric construction is itself not new: with zero thickness it is Cowan's iterated cell division [8], a member of the family of random tessellations that also contains the STIT model [9, 10]. Keeping both references in view, the recursive division and the local graph rule, turns most of the reported network statistics into exact or asymptotically exact statements.

The paper is organised as follows. Section II maps the SR construction onto Cowan's division and derives the statistics of the cell hierarchy. In Sec. III we write the dual growth as a face-local rule and obtain its degree distribution. Section IV computes the corner-cut probability by integral geometry and derives the clustering spectrum. Section V asks which properties survive when geometry is removed, and Sec. VI turns to distances and shortest-path multiplicity. Section VII identifies which rules move the degree exponent, and Sec. VIII discusses what this means for the reported structure–property relations.

II. THE SR CONSTRUCTION IS A RECURSIVE CELL DIVISION

We first set $\lambda_0 = 0$ and return to thickness at the end of this section. The current ligaments partition the square into convex polygonal cells. A uniform point falls in cell K with probability equal to its area A_K. Because K is convex and bounded by ligaments, the new ligament grown from that point stops on the boundary of K: it is the chord of K through a uniform point of K at a uniform angle. The SR model is therefore the iterated di-

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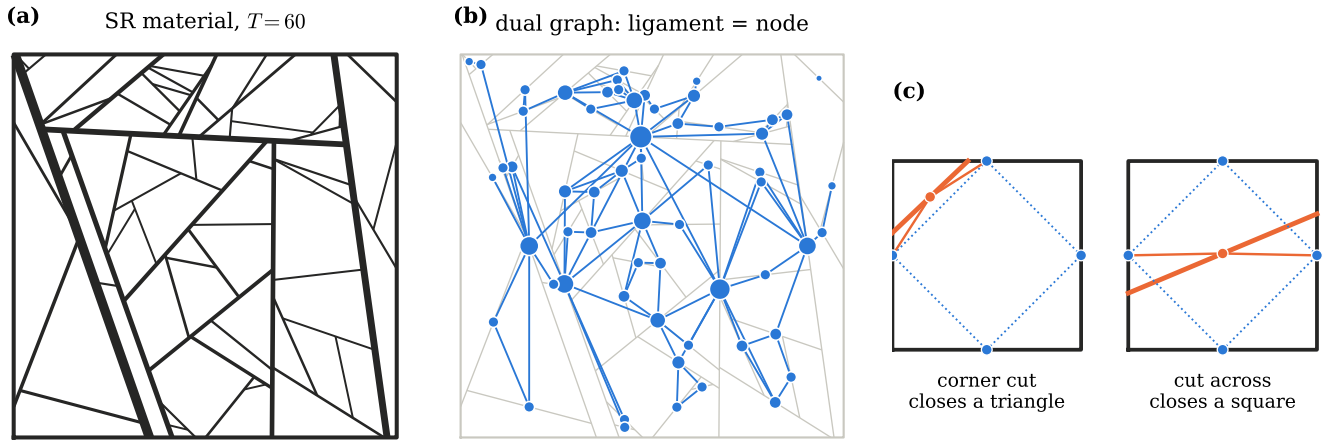


FIG. 1. (a) An SR realisation with $T = 60$ ligaments. The line width increases with $\lambda_t = \lambda_0 t^{-1/2}$. (b) Its dual graph: each ligament is a node, placed at the ligament midpoint with a size that grows with its degree, and each junction is a link. (c) A new ligament (orange) splits one cell. Dual nodes of the cell's sides are shown in blue, and the dotted lines join sides that meet at a corner. If the new ligament cuts off a corner, it joins two adjacent sides and closes a triangle in the dual. Otherwise it joins two sides at distance two and closes a square.

vision of cells with *area-weighted selection* and the *chord through a uniform point* division rule, model (a) of Fig. 2 in Cowan [8]. Cowan's rescaling of the window does not change the topology. STIT tessellations differ in both rules: cells are selected by perimeter and cut by isotropic uniform random lines [9, 10]. Gilbert's model [11], used as a reference architecture in Ref. [1], grows all segments simultaneously and is a different process.

Corner numbers. A junction placed on the side of a neighbouring cell is a straight angle there, so the corner number n of a cell is fixed at birth. A chord ending on two sides gives each daughter the two new corners at its ends, so $n_1 + n_2 = n + 4$. Starting from the square, after t steps there are $t+1$ cells and $4+4t$ corners, so the mean corner number is exactly $\langle n \rangle = 4$. Measured number fractions for $n = 3, \ldots, 7$ are $0.301, 0.452, 0.200, 0.042, 0.005$.

Uniform splitting and the binary search tree. Let K be a convex body, x a uniform point in K, and θ any direction independent of x. For fixed θ, the offset s of the line through x along the normal direction has density $w(s)/A_K$, where $w(s)$ is the chord length at offset s. The area on one side of the line is $\int_{-\infty}^{s} w(s')ds'$, so the area fraction is the cumulative distribution function of s evaluated at s. It is therefore uniform on $(0, 1)$, whatever the shape of K and the law of θ. Combined with area-weighted selection, this has a direct consequence. Label each cell by the interval of $[0, 1]$ that it would occupy if areas were laid on a line. The sequence of splits is then exactly that of a binary search tree (BST) built from uniform keys: the key falls in an interval with probability equal to its length and splits it at the key [12]. Hence the cell areas after T steps are the $T + 1$ spacings of T uniform points, $P(TA > x) \to e^{-x}$, which recovers Cowan's exponential area law [8], and $\sum_K A_K^2 = 2/(T + 2)$. The depth of the split at step t grows as $2 \ln t$, with variance $2 \ln t$, and areas never multiscale. Our simulations give a depth slope 1.983, a variance slope 1.980 and $\sum A^2 = 9.998 \times 10^{-5}$ at $T = 2 \times 10^4$ against 9.999×10^{-5} [Fig. 2(a,b)]. The multiscaling of Krapivsky and Ben-Naim [13] arises with rectangular cuts and lives in the lengths, not in the areas. Axis-aligned chords reproduce their width exponents $\gamma(a) = (a - \sqrt{a^2 + 4})/2$ within 2%, while with isotropic angles the perimeter moments scale as $t^{1-a/2}$, without multiscaling.

Lengths and the age of the host. Cells at time t have linear size $t^{-1/2}$, and the mean length of the ligament born at t is $\langle \ell_t \rangle \simeq c_\ell t^{-1/2}$ with $c_\ell = 1.16$. The total length grows as $L(t) \simeq c_L \sqrt{t}$ with $c_L = 2c_\ell = 2.33$. The first ligament has $\langle \ell_1 \rangle = 0.945$, the ℓ_0 of Ref. [1]. In the mean-field approximation of Sec. III, an end of ligament t lands on a ligament with probability proportional to its length, so the host's birth index h satisfies $P(h) \simeq \ell_h / L(t) \simeq \frac{1}{2}(ht)^{-1/2}$. Then $P(h \le xt) = \sqrt{x}$, that is

$$h/t \;\to\; U^2, \qquad U \sim \text{Uniform}(0, 1). \tag{1}$$

Equivalently, $\ln(t/h)$ is exponential with mean 2 [Fig. 2(c)]. Measured values of $\langle (h/t)^\alpha \rangle$ are 0.667, 0.498 and 0.330 for $\alpha = 1/4$, $1/2$ and 1, against $1/(1 + 2\alpha) = 0.667$, 0.500 and 0.333.

Thickness. With $\lambda_t = \lambda_0 t^{-\alpha}$, Eq. (1) gives the thickness ratio between a new ligament and its host as $\lambda_t/\lambda_h = U^{2\alpha}$. Its law does not depend on t, so every generation couples to the frame it attaches to in the same statistical way. Its moments $\langle (\lambda_h/\lambda_t)^p \rangle = 1/(1 - 2p\alpha)$ diverge at $\alpha = 1/(2p)$; for the bending stiffness of the host ($p = 3$) this happens at $\alpha = 1/6$. The fill fraction without overlaps is $\rho \simeq \lambda_0 \sum_t \ell_t t^{-\alpha} \simeq \lambda_0 c_\ell \sum_t t^{-\alpha - 1/2}$, which converges only for $\alpha > 1/2$. This recovers the boundary between the jammed and tunable-density phases of Ref. [1], and $\rho \simeq \lambda_0 c_\ell \zeta(\alpha + 1/2)$ matches their $\rho \approx \ell_0 \lambda_0 \zeta(\alpha + \beta)$

with $\beta = 1/2$. Thickness removes area and eventually stops growth in the jammed phase, but it does not change the rule by which cells are split (Sec. VII). In what follows the network is the zero-thickness one unless stated otherwise.

III. A FACE-LOCAL GROWTH RULE AND ITS DEGREE DISTRIBUTION

In the dual graph, step t reads as follows: choose a face (a cell) with probability equal to its area, choose the two sides of that face that the chord meets, and join the new node t to the two ligaments that carry them [Fig. 1(c)]. The new node never links to a node outside the chosen face. It is a local rule in the sense of Ref. [4], with faces in place of neighbourhoods.

Incidences equal degree. Call an incidence a pair (ligament i, cell K) in which i contributes a side to K. A new ligament has two incidences, one on each side, and two links. When a later ligament ends on i, it adds one link to i and cuts one of i's sides in two, adding one incidence. For a ligament whose ends are both interior, the number of incidences therefore equals the degree k_i at all times. Each step creates two hits and four incidences, so the total number of incidences is $4t$, in agreement with $\langle n\rangle = 4$.

Rate equation. If every incidence is equally likely to be hit, the rate at which node i gains links is $2k_i/4t$, linear preferential attachment with

$$\frac{dk_i}{dt} = \frac{k_i}{2t}, \qquad k_i(t) = 2\sqrt{t/i}. \tag{2}$$

This is the Barabási–Albert rule with $m = 2$ [14]. The master equation [15, 16] gives

$$p_k = \frac{12}{k(k+1)(k+2)}, \qquad P(k' \geq k) = \frac{6}{k(k+1)}, \tag{3}$$

for $k \geq 2$, so $\gamma = 3$. At $T = 10^5$ the measured $K^2P(k-2 \geq K)$ at $K = 8, 16, 32$ is $3.53, 4.45, 5.19$, against $3.49, 4.49, 5.16$ from Eq. (3) [Fig. 3(a)]. The degree of the cohort born at i decays as $i^{-\beta}$ with $\beta = 0.50$–0.51.

Metric form. The spatial preferential attachment of Ref. [1] is the same statement in metric form. Two ends land per step on a total length $L(t) = c_L\sqrt{t}$, so the hit rate per unit length is $2/L(t)$ and

$$k_i - 2 = \frac{4\ell_i}{c_L}\left(\sqrt{T} - \sqrt{i}\right). \tag{4}$$

The ratio of measured to predicted degree averages 1.006, with correlation 0.95 [Fig. 3(b)]. Equations (2) and (4) agree on average because $\langle \ell_i\rangle = c_\ell i^{-1/2}$ and $4c_\ell/c_L = 2$. They differ in where the randomness sits. In Eq. (2) it is in the history of hits. In Eq. (4) it is in the birth length, $x = \ell_i/\langle\ell_i\rangle$, with $\langle x^2\rangle = 1.39$. With $k \simeq 2x\sqrt{T/i}$ and i uniform, the tail amplitude is $K^2P(k \geq K) \to 4\langle x^2\rangle = 5.56$, about 7% below the value 6 of Eq. (3). A hub is therefore predictable at birth from its length, which is not the case in the Barabási–Albert model.

IV. CORNER CUTS, TRIANGLES AND CLUSTERING

A new node closes a triangle in the dual when the two ligaments it joins are already linked. Locally this happens when the chord cuts off a single corner of its cell, since the two sides meeting at that corner belong to linked ligaments.

Integral-geometry formula. Chords through a uniform point at a uniform angle sample lines with density $w(s)\,ds\,d\theta/(\pi A_K)$. For a fixed direction θ, order the corners by their projection on the normal. The chord cuts off the lowest corner alone when its offset lies between the projections of the lowest and second-lowest corners. The integral of $w(s)$ over that range is the area $A_{\rm lo}(\theta)$ cut off by the line through the second-lowest corner, and likewise $A_{\rm hi}(\theta)$ at the other end. Hence

$$P_{\rm corner}(K) = \frac{1}{\pi A_K}\int_0^\pi \left[A_{\rm lo}(\theta) + A_{\rm hi}(\theta)\right] d\theta. \tag{5}$$

For a triangle every chord cuts off a corner, so $P_{\rm corner} = 1$. For a square, Eq. (5) gives $2\ln 2/\pi = 0.4413$, for a 1×4 rectangle 0.303, for regular pentagons and hexagons 0.232 and 0.136 [17]. Averaged over the SR cell population, by corner number $n = 3, \ldots, 7$, the values are 1.000, 0.461, 0.254, 0.152 and 0.10, and the area-weighted average is 0.447–0.449. The measured frequency of corner cuts is 0.447–0.449.

Triangle bookkeeping. A corner cut creates a triangular daughter unless the parent was already a triangle, which can only split into $(3, 4)$. The number fraction of triangles is therefore $f_3 = \langle P_{\rm corner}\rangle - w_3$, where w_3 is the area fraction of triangles. Measured values are $0.30 = 0.449 - 0.147$.

Non-local closures. The fraction of new nodes that close a triangle is 0.496, not 1/2. Corner cuts account for 0.448. The other 0.048 join two non-adjacent sides of their cell whose ligaments already meet outside it. The remaining half of the new nodes close a square around the cell.

Clustering spectrum. Links between existing nodes are never added. A node therefore acquires triangles only at birth, with probability p_0, or when it is hit and the other end of the new ligament lands on one of its neighbours, with probability p per hit. The number of triangles at node i is $e_i = p_0 + p(k_i - 2)$, and

$$C(k) = \frac{2[p_0 + p(k-2)]}{k(k-1)} \simeq \frac{2p}{k}, \tag{6}$$

the hierarchical form produced by triadic closure [4, 5, 7]. With $p_0 = 0.494$ and a fitted $p = 0.53$, Eq. (6) is within 2% of the data for $k \geq 15$ [Fig. 3(c)]. The mean clustering

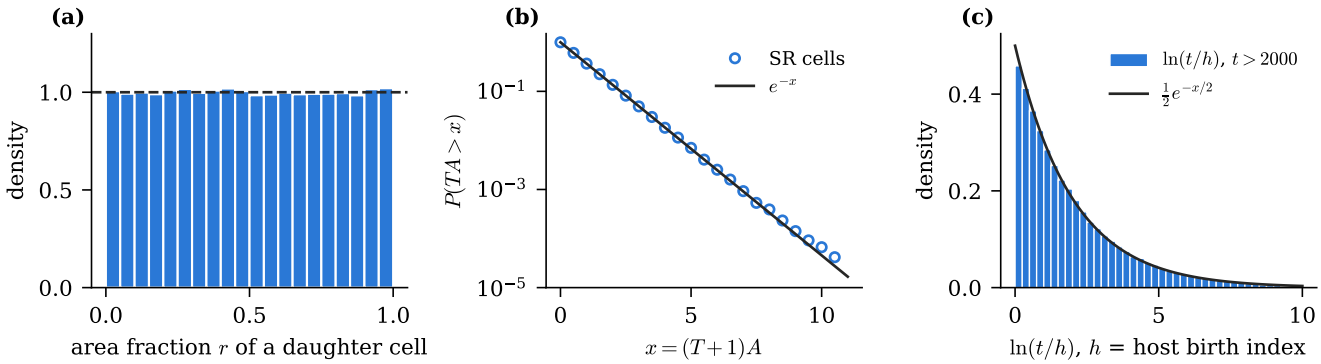


FIG. 2. The cell hierarchy of the SR construction (six realisations, $T = 2 \times 10^4$). (a) The area fraction of a daughter cell is uniform. (b) Rescaled cell areas follow e^{-x}, the spacing law of uniform points. (c) The birth index h of the ligament hit by a new ligament t satisfies $h/t \to U^2$, so $\ln(t/h)$ has density $\frac{1}{2}e^{-x/2}$.

is 0.38. At $k = 2$ the measured $C = 0.536$ exceeds p_0, because corner-cut ligaments are short and rarely hit again. The degree assortativity is $r \simeq -0.02$. The mean neighbour degree $k_{nn}(k)$ falls from 19.5 at $k = 2$ to a plateau of 11.1–11.5 for hubs, close to the Barabási–Albert value $(m/2) \ln T = \ln T = 11.5$ [Fig. 3(d)].

In the language of local rules, the SR dual mixes two moves. With probability $\simeq 0.45$ the new node attaches to both ends of a link (the corner), as in the edge-attachment model of Dorogovtsev, Mendes and Samukhin [6], except that the link is selected with probability proportional to the area of its cell rather than uniformly. With probability $\simeq 0.55$ it attaches to two nodes at distance two on a face cycle and closes a square. The edge-attachment model has $\gamma = 3$ and clustering 0.74 [6, 18]. The SR dual has the same γ and half the clustering; the other half of the closures went into squares. It is also the two-dimensional relative of the random Apollonian network [19, 20], in which faces are selected and a node's degree is proportional to the number of faces it bounds.

V. WHAT TOPOLOGY ALONE REPRODUCES

To separate the role of geometry, we grew a purely topological surrogate. Each face is stored as a cyclic list of the nodes that carry its sides. At each step a face is chosen uniformly. If it is a triangle, two of its sides are joined; otherwise two adjacent sides are joined with probability p_4, or two non-adjacent sides with probability $1-p_4$. The face is then split into two, with $n_1+n_2 = n+4$ as in the tessellation. No lengths or positions are used.

Every face-local variant we tried (faces chosen uniformly, by number of sides, or by a random-split area) reproduces Eq. (3): $\gamma_{\mathrm{MLE}}(k \geq 10) = 2.88$–$2.91$ and the same K^2P amplitudes. The single parameter $p_4 = 0.29$ also gives a mean clustering of 0.386 (SR, 0.377) and places $C(k)$ and $k_{nn}(k)$ on the SR curves. A variant in which sides are chosen with probability proportional to a mean-field length, instead of uniformly, breaks the equality between hits and incidences and gives $\gamma \simeq 3.6$. The degree statistics are therefore fixed by the face-local rule and by the equivalence of incidences, not by the geometry.

The surrogate fails on distances. At $T = 10^5$ its mean distance is 14.1 against 8.8 for SR, and it grows as $1.43 \ln T$ against $0.60 \ln T$ [Fig. 5(a)]. Faces cut across, in the surrogate, stack into long ladders of squares. In the geometric model chords hit long sides preferentially, and long sides belong to old, long ligaments, so new nodes stay close to the hubs and the dual remains small-world. Geometry matters for metric properties only.

VI. FEWEST-TURN PATHS AND THEIR MULTIPLICITY

A shortest path in the dual graph is a route through the material that changes ligament the fewest times. For a random graph with given degrees, the mean number of shortest paths between two nodes is $\langle \mu \rangle \simeq c_2/e$ with $c_2 = \langle k(k-1) \rangle / \langle k \rangle$, and local growth rules depart from this value in a rule-dependent way [21]. For $\gamma = 3$, c_2 grows as $\ln T$; we measure $c_2 = 7.9$ at $T = 10^3$ and 15.6 at $T = 3 \times 10^5$.

The degree-preserving randomisation of the SR dual follows the configuration-model value, with $\langle \mu \rangle / (c_2/e) = 1.03$, 0.98, 0.95, 0.94 and 0.89 for $T = 10^3, \ldots, 10^5$. The SR dual itself exceeds it by a factor that grows with size: 2.3, 2.8, 3.9, 4.9, 4.9 and 6.8 for $T = 10^3, \ldots, 3 \times 10^5$. The edge-attachment model stays close to the baseline (0.9–1.2) over the same range, so triangles add no redundancy. The excess comes from the square-closing half of the splits: the number of shortest paths grows with distance at a rate of 0.45–0.6 per hop, as on a square lattice, where fewest-turn routes through a grid of ligaments are many. Because μ has a heavy tail, Fig. 5(b) shows $\langle \ln \mu \rangle$,

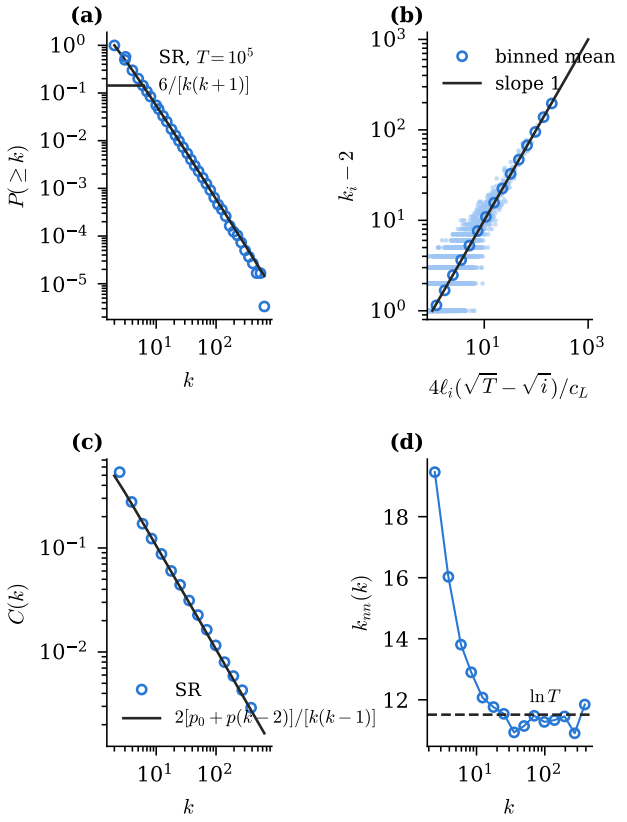


FIG. 3. Dual graph of the SR construction, three realisations with $T = 10^5$. (a) Cumulative degree distribution and Eq. (3). (b) Degree against the prediction (4) from birth length and age (light dots: a random sample of ligaments; circles: binned means). (c) Clustering spectrum and Eq. (6) with $p_0 = 0.494$, $p = 0.53$. (d) Mean neighbour degree; the dashed line is $\ln T$.

which grows as $0.18 \ln T$ for SR against $0.06 \ln T$ for the randomised graph. In the surrogate the ladders make the multiplicity explode ($\langle \ln \mu \rangle = 3.4$ at $T = 3 \times 10^5$). The squares supply the redundancy, while the geometry keeps them from stacking.

VII. WHAT MOVES THE DEGREE EXPONENT

In a growth rule where each new node creates a fixed number of incidences and old nodes are hit in proportion to theirs, the exponent is $\gamma = 1 + 1/b$, with b the number of links received by old nodes per step divided by the number of incidences created per step. In two dimensions every new ligament makes two junctions and four incidences, so $b = 1/2$ and $\gamma = 3$, whatever the cell selection, the angle and the thickness, provided the hit rate per incidence is uniform across nodes.

We tested this with the design rules of Ref. [1] at $T = 3 \times 10^4$, two realisations each [Fig. 4(a)]:

- Thickness in the tunable-density phase ($\alpha = 0.5$–1, $\lambda_0 = 0.01$–0.1, $\rho = 0.12$–0.45) gives $\gamma_{\mathrm{MLE}} = 2.85$–$2.92$, $\beta = 0.50$–0.52, $C = 0.376$–0.378, and a triangle closure of 0.49–0.50.
- In the jammed phase ($\alpha = 0$–0.4), growth stops when the free area is exhausted (at $t \simeq 2000$ for $\alpha = 0$, $\lambda_0 = 0.02$). Jamming truncates T; it does not change γ (2.89–2.94 wherever the statistics allow) or C (0.35–0.37).
- Angle sets change clustering but not γ. With angles $\{0, \pi/2\}$ all cells are rectangles, so $C = 0$ exactly and $\gamma = 2.87$; this is the rectangular fragmentation of Ref. [13]. With $\{0, \pi/3, 2\pi/3\}$, $C = 0.17$.
- Global alignment (von Mises, κ up to 16), cell selection proportional to area^q ($q = 0, 0.5, 2$), and linear or radial density gradients give $\gamma = 2.86$–2.96. Selection with $q = 0$ raises C to 0.45, and $q = 2$ lowers it to 0.35.

The angle set and the selection weight therefore tune clustering between 0 and 0.45, while the exponent stays at 3.

Coherent orientation fields. The polarized rule of Ref. [1], in which the angle follows a prescribed field $\theta(x, y)$, breaks the uniformity of the hit rate. Locally parallel newcomers cannot hit each other; they end on the ligaments that are misaligned with the local field, and for straight ligaments in a curved field the misalignment grows with length. The degree then grows faster than length, $k \sim \ell^{1.5\text{–}1.6}$ instead of $\ell^{1.03}$. For a circular field and for waves $\theta = \pi(x + y)/w$ with $w = 0.125$–1 we find $\gamma \simeq 2.44$–2.51 and cohort exponents $\beta \simeq 0.6$–0.67, stable from $T = 10^4$ to 2×10^5. At $T = 10^6$ the maximum-likelihood exponent is 2.43–2.50 ($k \geq 10$ to $k \geq 40$) for the circular field and 2.36–2.47 for the shortest wave ($w = 0.125$), while the uniform control gives 2.88–2.98, so the effect does not wash out at this size. The exponent decreases continuously as the fraction of ligaments following the field increases [Fig. 4(b)].

Three dimensions. For plates in a cube, a new plate meets $\bar{m}$ existing plates (the sides of its section polygon), creates $2+\bar{m}$ incidences and hands out $\bar{m}$ links. Then $b = \bar{m}/(2+\bar{m})$ and $\gamma = 2+2/\bar{m}$. We measure $\bar{m} = 4.47$–4.57, which predicts $\gamma = 2.44$. The dimensional argument $k \sim (\ell/\ell_T)^{d-1}$ gives $\gamma = (2d-1)/(d-1) = 2.5$, the exponent of random Apollonian networks in d dimensions [22]. At $T = 8 \times 10^4$ the measured value is 2.50 ($k \geq 40$), with $\beta = 0.66$. Three-dimensional SR materials therefore have $\gamma \simeq 2.5$, not 3.

VIII. CONSEQUENCES FOR STRUCTURE–PROPERTY RELATIONS

At fixed α, the thickness of ligament i is $\lambda_0 i^{-\alpha}$ exactly, and its degree and length both scale as $i^{-1/2}$ on average [Eqs. (2) and (4)]. The marginal laws reported in Ref. [1], an energy that decays as $i^{-(3/4+\alpha/2)}$ with

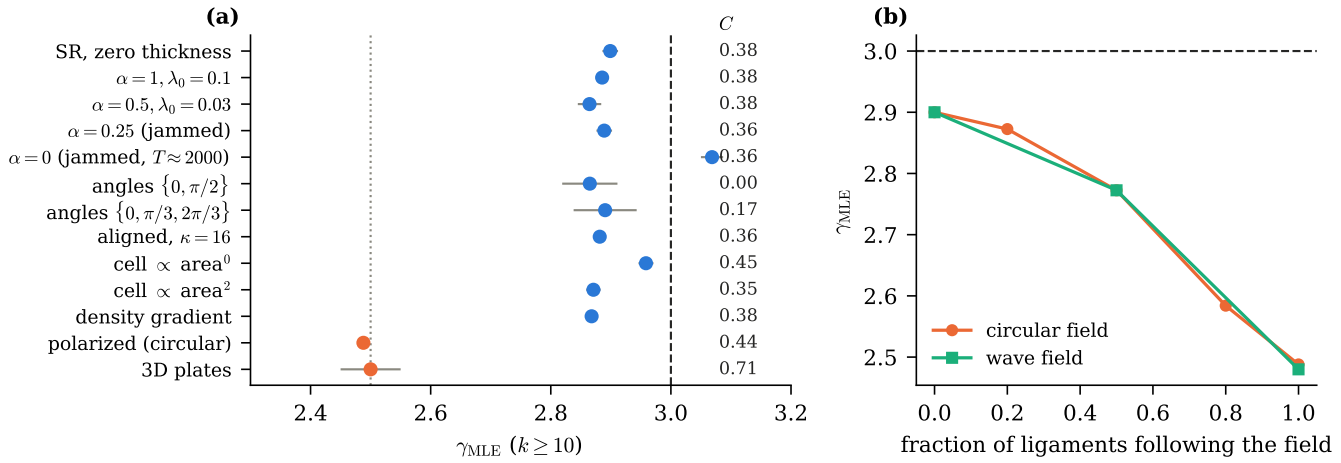


FIG. 4. (a) Maximum-likelihood exponent ($k \geq 10$) for the design rules of Ref. [1], $T = 3 \times 10^4$, two realisations (the jammed $\alpha = 0$ case stops at $T \simeq 2000$; 3D plates at $T = 8 \times 10^4$, $k \geq 40$). The number on the right is the mean clustering coefficient. Dashed line: $\gamma = 3$; dotted line: $\gamma = 2.5$. (b) Exponent against the fraction of ligaments whose angle follows a circular or a wave orientation field ($T = 3 \times 10^4$); the remaining ligaments have uniform angles.

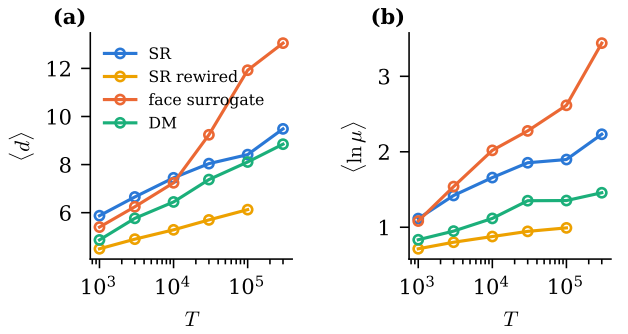


FIG. 5. (a) Mean distance in the dual graph against T for the SR construction, its degree-preserving randomisation, the topological face surrogate ($p_4 = 0.29$) and the edge-attachment (DM) model [6]. (b) Mean logarithm of the shortest-path multiplicity. Averages over three realisations and 100 sources.

the introduction time and the corresponding powers of k, ℓ and λ, are one age law written in different variables. A product $U \sim k^a \lambda^b$ reproduces the age law for every α only if $a/2 + \alpha b = 3/4 + \alpha/2$, that is $a = 3/2$ and $b = 1/2$, the reported exponents. Because $k \propto \ell$ at fixed age [Eq. (4)], the same data are described equally well by $U \sim \ell^{3/2} \lambda^{1/2} f(\theta)$. Degree, length and age cannot be separated from cohort scaling alone.

As an illustration, we computed per-ligament energies in an Euler–Bernoulli frame (axial area λ, bending moment of inertia $\lambda^3/12$, rigid T-junctions, affine uniaxial strain on the boundary) for $T = 1000$–5000. Fits on degree and on length are indistinguishable, $U \propto k^{1.64} \lambda^{1.61}$ ($R^2 = 0.634$) against $U \propto \ell^{0.82} \lambda^{2.02}$ ($R^2 = 0.631$), and the age alone gives an R^2 within 0.03–0.08 of the best fit. Within a birth cohort the partial correlations of U with k (0.22–0.30) and with ℓ (0.17–0.34) are comparable. The thickness exponent of a slender-beam frame lies between stretching (1) and bending (3) and is not 1/2. A frame of slender beams is not the continuum finite-element model of Ref. [1], so we do not attach weight to this exponent. The identifiability statement does not depend on it.

The same reasoning applies to design. Thickness is a function of birth order, and degree and length are statistically slaved to birth order, so α and λ_0 act on the material through the age of each ligament on a fixed network. Connectivity becomes a design variable only when the growth rule is changed so that b moves away from 1/2, as with coherent orientation fields or in three dimensions.

IX. DISCUSSION

The scale-rich construction joins two known structures. Geometrically it is Cowan's recursive division of cells [8]; as a graph it is a face-local growth rule [4] whose degree distribution is the Barabási–Albert form with $m = 2$. The recursion explains the cell hierarchy: uniform split fractions, a binary-search-tree genealogy and exponential areas. The local rule explains the network: linear preferential attachment from the identity between degree and face incidences, a hierarchical clustering spectrum from corner cuts, and an excess of shortest paths from square-closing splits. Keeping both references in view settles which features are fixed by the class and which can be designed. The exponent $\gamma = 3$, the equality $\langle n \rangle = 4$, the host-age law $h/t \to U^2$ and the stationarity of the thickness ratio are fixed. Clustering is designable through the angle set and the selection weight. The degree exponent

is designable only through rules that change b.

Several points remain open. The uniformity of the hit rate per incidence is a mean-field assumption, accurate to a few per cent in the tail but not proven. The per-hit triangle probability $p \simeq 0.53$ and the non-local closure fraction 0.048 require the joint law of cell shapes next to long ligaments. Under coherent orientation fields we observe $\gamma \simeq 2.4$–2.5 up to $T = 10^6$ but have no theory of its value. In three dimensions the two estimates, 2.44 and 2.5, are not separated by our data. Finally, the growth of the shortest-path multiplicity with T is consistent with both a log-quadratic and a weak power law over the range studied.

More broadly, the SR dual is a reminder that a spatial growth process can carry a non-spatial growth class. Here the area-weighted point and the convexity of the cells do the work of a random walk on the graph: they deliver the new node to a face, and the face does the rest.

ACKNOWLEDGMENTS

All code is available at https://github.com/av2atgh/scalerich. The calculations were assisted by Claude Opus 5, an AI assistant developed by Anthropic. Nodes & Links Ltd provided support in the form of salary for Alexei Vazquez but did not have any additional role in the conceptualization of the study, the analysis, the decision to publish or the preparation of the manuscript.

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